Deep Learning for Wagner-Grade Classification of Diabetic Foot Ulcers: ResNet50 vs. Vision Transformer

Chengxuan Xia
Department of Biomedical Engineering
Carnegie Mellon University
chengxux@andrew.cmu.edu

Abstract

Diabetic foot ulcers (DFU) are a leading cause of lower-limb amputation, and timely Wagner-grade staging from clinical photographs could improve triage in low-resource settings. We frame DFU staging as a 5-class image classification problem on a dataset of 3,886 deduplicated photographs and compare two ImageNet-pretrained backbones with the same training pipeline: a ResNet50 baseline (CNN with hierarchical local receptive fields) and a ViT-B/16 variant (pure self-attention over 16×16 patches). Both models reach roughly the same test-set performance (ResNet50: 81.65% accuracy / 0.753 macro-F1; ViT-B/16: 81.48% / 0.747), and both make their errors on the same clinically ambiguous boundaries (Grade 1$\leftrightarrow$2 and Grade 2$\leftrightarrow$3). The result suggests that, on this dataset and at this scale, the choice of inductive bias matters less than the underlying class ambiguity in Wagner staging from photographs alone.

1 Introduction

A diabetic foot ulcer (DFU) is an open wound on the foot of a person with diabetes, caused by a combination of peripheral neuropathy (loss of protective sensation) and peripheral arterial disease (impaired healing). Untreated, DFUs progress from superficial skin breaks to deep tissue infection and gangrene, and they precede roughly 80% of non-traumatic lower-limb amputations. Clinical management depends critically on *stage*, and the most widely used staging scheme is the Wagner Ulcer Classification System, which assigns a grade from 0 (intact, pre-ulcerative skin) through 5 (gangrene of the entire foot). The grade determines whether a patient can be managed conservatively with offloading and dressings (Grade 0–1), needs aggressive debridement and antibiotics (Grade 2–3), or requires urgent surgical referral (Grade 4–5).

Wagner grading is currently performed in person by a trained clinician. In rural and primary-care settings without a wound specialist on staff, this creates a triage bottleneck: patients with high-grade ulcers may be missed during routine check-ups, while patients with low-grade ulcers may be unnecessarily referred. A model that maps a single smartphone photograph of the foot to a Wagner grade would let a primary-care provider, a community health worker, or even the patient themselves obtain a first-pass severity estimate, prioritising specialist time toward the cases that need it most. This is a natural fit for deep learning: the visual cues that define each Wagner grade (superficial redness, exposed dermis, depth of the wound bed, presence of necrotic tissue) are appearance-based and have been shown to be learnable from a few thousand labeled images. The goal of this project is therefore to build and analyze a Wagner-grade classifier from RGB photographs alone, and to study how the choice of vision backbone affects the kinds of errors the model makes.

Preprint. Under review.

2 Methods

Dataset and pre-processing. We use a curated dataset of 3,887 RGB diabetic foot photographs assembled from a combination of web-scraped clinical images and open sourced clinical datasets, all manually labeled with Wagner grades 0 through 5 by Rubitection (Medical technology start-up with a focus on AI powered wound staging). Because our dataset contains relatively few Grade 5 examples, and because Grades 4 and 5 share the same underlying clinical sign (necrosis, distinguished only by extent — localised in Grade 4 vs. widespread in Grade 5), we merge the two into a combined "Grade 4–5" class for reporting purposes After MD5-based deduplication (1 duplicate removed), 3,886 images remain. The class distribution is heavily imbalanced: Grade 0 (1,237), Grade 1 (479), Grade 2 (1,633), Grade 3 (287), Grade 4–5 (250). We split the data into 70% train / 15% validation / 15% test (2,720 / 583 / 583) using stratified sampling on the label so that the class proportions are preserved across splits to within 0.1 percentage points. All images are resized to 256 px on the short side and center- or random-cropped to 224×224. The training pipeline uses random resized crops (scale 0.7–1.0), horizontal and vertical flips (a foot photo has no canonical orientation), $\pm 30°$ rotation, color jitter (brightness/contrast/saturation/hue), small affine translation and scaling, and Random Erasing with $p = 0.25$ to simulate occlusion by dressings or shadows. To counteract the class imbalance during training, we draw mini-batches with a `WeightedRandomSampler` whose per-sample weight is the inverse of its class frequency, giving an approximately balanced class distribution within each batch (verified empirically: ≈ 320 samples per class over 50 batches). Validation and test loaders preserve the original imbalance, so reported test metrics reflect realistic class frequencies. We optimise cross-entropy with label smoothing $\varepsilon = 0.1$ using AdamW, a cosine learning rate schedule over 30 epochs, and early stopping with patience 8 on validation loss.

Baseline model: ResNet50. ResNet50 [1] is the canonical CNN baseline covered in the course and is a natural fit for a medical-image classification task at this scale. The model is a 50-layer residual network composed of 3×3 convolutions organised into bottleneck blocks, with skip connections that allow gradients to bypass layers and enable deep optimization. Two properties make it well-suited to DFU images. First, its strong locality bias (each output unit sees a bounded receptive field of nearby pixels) matches the fact that Wagner grade is determined by local texture and color cues at the wound bed, not by long-range relationships across the image. Second, its translation equivariance (a wound shifted to a different part of the foot produces the same features, just shifted) is exactly the invariance we want when the same lesion can appear anywhere in the frame. We initialise from ImageNet-1k pretrained weights, replace the final 1000-way classifier with a Dropout($p = 0.3$) $\rightarrow$ Linear(2048, 5) head, and fine-tune the entire network with a single learning rate of 5×10^{-5} and weight decay 10^{-3}.

Variant: Vision Transformer (ViT-B/16). The variant is the ViT-B/16 architecture of Dosovitskiy et al. [2]. ViT divides the input image into a grid of non-overlapping 16×16 patches (196 patches at 224×224), linearly projects each patch into a 768-dimensional token, prepends a learnable [CLS] token, adds a learned position embedding, and processes the sequence with 12 layers of multi-head self-attention. Classification reads out the [CLS] token through a linear head. The choice is motivated by a hypothesis specific to Wagner grading: while a low-grade ulcer can be diagnosed from a single localised region (the wound itself), higher Wagner grades depend on *relationships* between regions of the foot — e.g., distinguishing localised gangrene of one toe (Grade 4) from gangrene of the whole foot (Grade 5) requires comparing tissue color across spatially distant regions. ViT's global self-attention from the first layer onward should, in principle, capture such inter-region relationships more directly than a CNN. ViT lacks the locality and translation-equivariance priors of a CNN, so we expect it to be more sample-hungry; we partially compensate by initialising from ImageNet-1k pretrained weights and using a layer-wise learning rate (5×10^{-4} on the new classification head, 5×10^{-5} on the pretrained backbone) with weight decay 0.05, following common practice for fine-tuning transformers.

3 Results and Discussion

Metrics. We report top-1 accuracy and macro-averaged F1. Accuracy reflects the realistic class imbalance of the test set, while macro-F1 averages per-class F1 with equal weight and so penalises a model that does well only on the majority classes (Grade 0, Grade 2). We also report per-class precision/recall/F1 and the confusion matrix, because the clinical cost of confusing nearby grades (e.g., Grade 1 vs. Grade 2) is much smaller than the cost of confusing distant grades (e.g., Grade 0 vs. Grade 4–5).

Table 1: Test-set performance on 583 held-out DFU images (5-way Wagner classification).

Model	Accuracy	Macro-F1
ResNet50 (baseline)	0.8165	0.7530
ViT-B/16 (variant)	0.8148	0.7468

Table 2: Per-class F1 on the test set. Support (number of test images): 186 / 72 / 245 / 43 / 37.

Model	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4–5
ResNet50	0.958	0.723	0.801	0.533	0.750
ViT-B/16	0.949	0.699	0.815	0.495	0.775

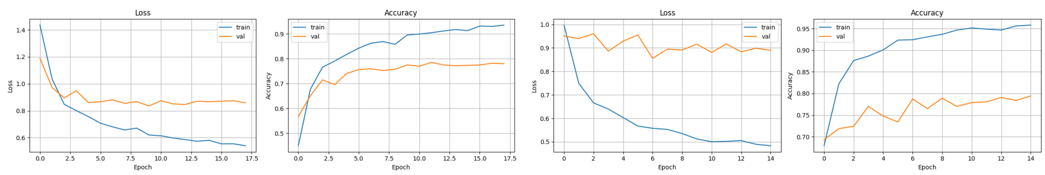

Figure 1: Training and validation curves. Left: ResNet50, early-stopped at epoch 18 with best validation loss 0.835. Right: ViT-B/16, early-stopped at epoch 15 with best validation loss 0.855. Both models show a clear train–val generalisation gap that opens after $\sim$5 epochs and never closes, indicating that the bottleneck on this dataset is data, not optimization.

Quantitative comparison. Table 1 shows that the two models reach essentially the same top-line performance: ResNet50 is ahead by 0.17 points of accuracy and 0.62 points of macro-F1, well within the noise of a single training run on a 583-image test set. Table 2 confirms that the per-class profile is also similar, with both models scoring above 0.94 on the easy Grade 0 class and below 0.55 on the hardest Grade 3 class. The two models trade small amounts of per-class F1: ResNet50 is slightly stronger on Grade 1 and Grade 3, while ViT is slightly stronger on Grade 2 and Grade 4–5.

Where the errors actually live. The confusion matrices in Figure 2 are more informative than the headline numbers. For both architectures, errors cluster on the off-diagonal ± 1 band, i.e. adjacent Wagner grades. The dominant failure mode in both models is Grade 2 being absorbed into Grade 1 (25 images for ResNet50, 14 for ViT) and into Grade 3 (22 for ResNet50, 29 for ViT), reflecting genuine clinical ambiguity around the boundary at which an ulcer is judged to penetrate to deeper tissue. Crucially, far-off-diagonal errors — the clinically dangerous ones, e.g., calling a Grade 4–5 wound a Grade 0 — are essentially absent in both models (ResNet50: 1/37; ViT: 0/37). This is the kind of error pattern a triage tool should produce: it sometimes misranks neighbouring grades, but very rarely misses a severe case entirely.

Was the hypothesis supported? Our hypothesis was that ViT's global self-attention would help on the higher Wagner grades, where the distinction depends on relationships between spatially distant regions of the foot. The data do not clearly support this hypothesis. ViT does outperform ResNet50 on Grade 4–5 (F1 0.775 vs. 0.750), which is consistent with the prediction, but the absolute gap is small and ViT *underperforms* ResNet50 on Grade 3 (F1 0.495 vs. 0.533), which is the other grade where global context should arguably help.

4 Conclusion

We trained an ImageNet-pretrained ResNet50 and an ImageNet-pretrained ViT-B/16 with an identical data pipeline, augmentation policy, class-rebalancing strategy, and training schedule on 2,720 Wagner-graded DFU photographs. On the held-out test set the two models perform nearly identically (81.6%

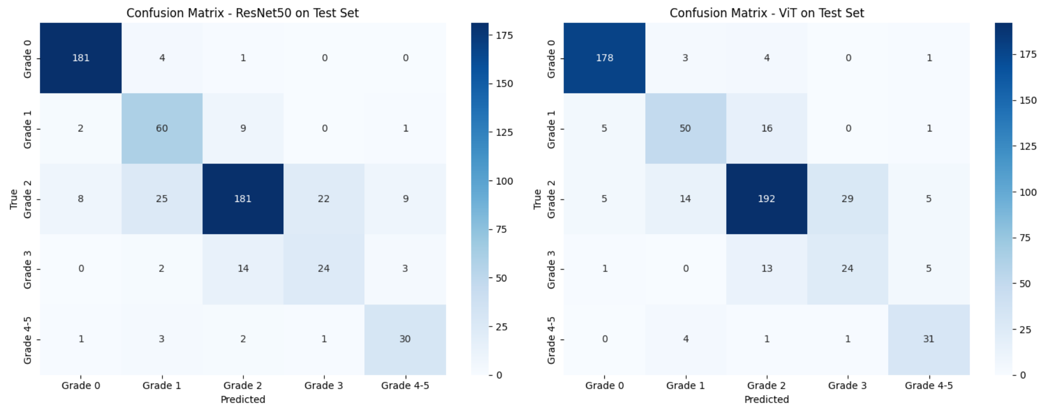

Figure 2: Test-set confusion matrices. Left: ResNet50. Right: ViT-B/16. Both models concentrate their errors on adjacent-grade confusions (the diagonal ± 1 band), particularly Grade 2 being misclassified as Grade 1 or Grade 3.

vs. 81.5% accuracy; 0.753 vs. 0.747 macro-F1) and, more interestingly, fail in the same way: both confuse adjacent Wagner grades, particularly around the Grade 2 boundary, while reliably separating clinically distant grades. Our hypothesis that ViT's global attention would meaningfully help the higher grades was not supported — the per-class differences between the two backbones are within run-to-run noise, and the curves indicate that the binding constraint is generalisation from a few thousand images on a problem with genuinely ambiguous boundaries, not the choice of inductive bias.

The most surprising part of the result is how cleanly the two models converge to the same error pattern even though their inductive biases (locality + translation equivariance vs. global self-attention) are nominally very different. This suggests that the dataset itself, not the architecture, sets the ceiling.

Given more time and compute, the most promising next steps are: (1) collecting or labeling more Grade 3 and Grade 4–5 images, since both models bottleneck on these minority classes; (2) reformulating the problem as ordinal regression rather than nominal classification, so that an off-by-one error is penalised less than an off-by-three error during training; (3) test-time augmentation, which is essentially free and tends to recover 1–2 points of accuracy on small medical-imaging datasets; and (4) an interpretability analysis (Grad-CAM for ResNet50, attention rollout for ViT) to verify that both models are attending to the wound itself rather than to spurious cues such as skin tone, dressing material, or background.

Collaboration Statement

This is a solo project; all model implementation, training, evaluation, and writing were done by the author. Claude was used in two ways. First, during code development they were used as a programming assistant to debug PyTorch errors, look up `torchvision` API details, sanity check code blocks, and graphing code blocks. Second, this report is an AI revised version of the author's own writing and audio transcription; all technical claims, hyperparameter choices, results, and the analysis of confusion patterns are the author's own, and every AI-suggested sentence was reviewed and edited before inclusion. No AI tool produced any of the reported numbers; all metrics in this document were generated by running the attached notebooks.

References

[1] K. He, X. Zhang, S. Ren, and J. Sun. Deep Residual Learning for Image Recognition. In *CVPR*, 2016.

[2] A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *ICLR*, 2021.